

The Price of Irreversible Representation Growth

Sharp Four-State Obstructions under Logarithmic Loss

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Abstract

Can increasingly informative representations be obtained by refining their existing structure while remaining competitive at every capacity? We study this question through the best prediction loss attainable with a bounded number of representation states. A full-support family with four source states and a binary target has minimum simultaneous log-loss ratio $\sqrt{L}/(2\sqrt{\log 2})(1 + o(1))$ over all deterministic hierarchies, where the least source probability is e^{-L} . Allowing arbitrary stochastic Markov refinements improves the sharp constant by a factor of two, but the ratio remains unbounded. The conclusions concern total Bayes logarithmic loss, hold with complete knowledge of the distribution, and use the smallest possible nontrivial source and target cardinalities. We prove the deterministic formula by enumerating all partitions and obtain the stochastic converse through a finite random-function representation. We also prove a $10(2 + \log(1/\gamma))$ upper bound for arbitrary four-state sources whose masses are at least γ , and an excess-loss extension for convex generators with divergent endpoint slope. The qualitative four-point merge conflict is known; the contribution is its sharp, unbounded all-hierarchy and stochastic quantification, together with explicit regularity and reassignment boundaries. Reproducible high-precision calculations evaluate every deterministic hierarchy and bracket the stochastic optimum.

Contents

1	Motivation and contribution	2
2	Closest prior work and the novelty boundary	2
3	Statistical setting and admissible structural changes	3
4	A sharp four-state obstruction	4
5	Randomized hierarchies and stochastic refinements	7
6	Regularity, robustness, and the cost of reassignment	8
7	An extension to convex Bayes-risk generators	11
8	Reproducible computational investigations	13
9	Interpretation and remaining questions	16

1 Motivation and contribution

Information can describe what a representation preserves without determining whether a sequence of structural changes can realize the best representation at each capacity. A system that only divides existing states has a particular constraint: every later state must belong to one earlier state. Even an oracle that knows the full task distribution cannot reverse an incompatible earlier grouping through division alone. This distinction suggests studying the *price of irreversible growth*, rather than postulating that high activation entropy should trigger neuron splitting.

We make the question precise for finite representations of a source X used to predict a target Y . A representation with at most k states incurs Bayes logarithmic loss $H(Y | Z)$. The flat optimum R_k^* can redesign its encoder independently for every k . An irreversible growth path must instead make its representations compatible: the coarser representation is a function, or a stochastic degradation, of the finer one. This is an abstraction of changes in representational organization. It does not identify a representation state with a biological neuron, impose a specific neural parameterization, or analyze gradient descent.

The principal result is stronger than the familiar observation that capacity-specific optima need not be nested. For an explicit four-state family, *every* deterministic hierarchy has an arbitrarily large multiplicative loss at either two or three states. Optimally chosen stochastic encoders and stochastic coarsening do not remove this obstruction. Their optimal asymptotic price is exactly half the deterministic price. These are population statements about the admissible representations, with no restriction on computation, sample size, or the information available to the structural controller.

The constructive mechanism involves two common nearly deterministic inputs and two rare intermediate inputs. At two states, the rare inputs join their respective nearby anchors. At three states, the optimal representation separates the anchors and groups the rare inputs together. The input weights, posterior probabilities, and noise scale are chosen so that *both* possible ways to preserve a hierarchy become arbitrarily expensive relative to the appropriate flat optimum. Irreducible label noise is smaller than the relevant optimal risks, so the result applies to total prediction loss.

The conclusions also have precise boundaries. A lower bound on source probabilities gives a uniform upper bound in the four-state problem, without a lower bound on label probabilities. Permitting reassignment of one rare input, of mass e^{-L} , connects the two exact optima. A general convex-generator theorem extends the excess-loss obstruction beyond Shannon entropy. We give full proofs, including the elementary comparison inequalities used in the positive result.

2 Closest prior work and the novelty boundary

Agglomerative information bottleneck (AIB) already optimizes task information under hard partition cardinality and chooses merges by their information loss [8]. The extended report explicitly discusses the failure of optimal partitions to form a hierarchy [7, Section 3.1]. The objective and that qualitative observation are therefore established starting points, not contributions of this manuscript.

The closest concrete precursor is Kartowsky and Tal [6, Section III-A, Example 1]. Their binary-input, four-output channel corresponds, after reversing the probabilistic roles, to four equally weighted posterior points 0, 1/3, 2/3, 1. A middle-pair merge conflicts with the best subsequent two-cell partition. Their main result gives an additive worst-case power law for greedy degradation as the output budget grows. Our question instead compares *every* compatible path with the flat optimum on the *same instance* at two fixed budgets. The

rare-mass construction proves an unbounded multiplicative ratio, extends it to total log-loss with full support, and resolves its stochastic asymptotic constant. The underlying four-point pattern itself is not new.

The simultaneous approximation criterion is the price-of-hierarchy criterion from hierarchical clustering; see Arutyunova and Röglin [1, Introduction and Section 2]. Their sharp results concern radius and diameter in metric spaces. KL centroid clustering belongs to the established Bregman framework [2, Sections 2–3]. Chaudhuri and McGregor [4, Theorem 8 and Lemmas 12 and 18] compare KL clustering with Hellinger geometry and obtain logarithmic approximation bounds for flat clustering. Their cluster-cost and bounded likelihood-ratio comparisons motivate our source-mass argument; we prove the required weighted inequalities and four-point hierarchy bound explicitly.

Charvin, Catenacci Volpi, and Polani [3, Definition 5, Proposition 4, and Sections 2.3 and 3.2] study exact and soft successive refinement of the stochastic information bottleneck under information-rate constraints. Their compatibility conditions and examples make nonnesting a known issue in that setting as well. A rate constraint $I(X; Z) \leq R$ and a cardinality constraint $|\mathcal{Z}| \leq k$ are different resources. Our result uses the latter and quantifies the best possible simultaneous multiplicative loss under hard alphabet budgets, a different resource from their mutual-information tradeoff.

The novelty assessment is limited to the primary sources and targeted searches recorded with this project. No equivalent sharp all-hierarchy four-state log-loss theorem, or its stochastic constant, was found in that search. The present claim of contribution is the precise theorem package below, not a claim to have discovered KL clustering, successive refinement, or the original merge conflict.

3 Statistical setting and admissible structural changes

Let X have a finite alphabet $\mathcal{X}$, positive masses p_x , and a finite target alphabet $\mathcal{Y}$. Write $P_x = P(Y = \cdot \mid X = x)$. All logarithms are natural, $0 \log 0 = 0$, and all entropies are finite. A deterministic representation is a partition $\mathcal{P}$ of $\mathcal{X}$; its output identifies the cell containing X . For a cell C , put

$$p_C = \sum_{x \in C} p_x, \quad P_C = p_C^{-1} \sum_{x \in C} p_x P_x.$$

Predicting P_C minimizes expected logarithmic loss conditional on C : for any prediction q , its conditional expected loss equals $H(P_C) + D_{\text{KL}}(P_C \parallel q)$. Thus

$$R(\mathcal{P}) = H(Y \mid \mathcal{P}(X)) = \sum_C p_C H(P_C), \quad (1)$$

$$D(\mathcal{P}) = R(\mathcal{P}) - B_0 = \sum_C \sum_{x \in C} p_x D_{\text{KL}}(P_x \parallel P_C), \quad B_0 = H(Y \mid X). \quad (2)$$

The second identity follows by expanding the logarithms and using the cell-mean formula. It also gives $D(\mathcal{P}) = I(X; Y) - I(\mathcal{P}(X); Y)$. Hence minimizing total or excess log-loss selects the same flat partitions, but their approximation ratios are different when $B_0 > 0$.

Let R_k^* and D_k^* be the minimum risks over partitions into k nonempty cells, for $1 \leq k \leq |\mathcal{X}|$. Allowing at most k cells gives the same value, since splitting a cell cannot increase conditional entropy. A deterministic hierarchy $(\mathcal{H}_k)_{k=1}^n$ has exactly k cells at level k and $\mathcal{H}_{k+1}$ refines $\mathcal{H}_k$. Each step is a binary split in the growth direction. For $n = 4$ define

$$\rho_{\text{det}} = \min_{\mathcal{H}} \max_{k \in \{2,3\}} \frac{R(\mathcal{H}_k)}{R_k^*}. \quad (3)$$

Our full-support examples have positive denominators. Levels one and four have fixed representations and ratio one, so omitting them changes nothing.

A stochastic refinement consists of channels $K_3 : X \rightarrow Z_3$ and $G : Z_3 \rightarrow Z_2$, with at most three and two output states respectively. The joint law is

$$p(x, y, z_3, z_2) = p(x, y)K_3(z_3 | x)G(z_2 | z_3), \quad (4)$$

so $Y - X - Z_3 - Z_2$ is a Markov chain. Define

$$\rho_{\text{stoch}} = \inf_{K_3, G} \max \left\{ \frac{H(Y | Z_2)}{R_2^*}, \frac{H(Y | Z_3)}{R_3^*} \right\}. \quad (5)$$

This includes all deterministic hierarchies. No external random variable is provided to the predictor for free in (5).

Lemma 3.1 (Flat stochastic encoders do not improve the benchmark). *For finite X, Y and a fixed output budget k , the minimum of $H(Y | Z)$ over all channels $X \rightarrow Z$ with at most k states is R_k^* .*

Proof. Every channel is a mixture of deterministic function tables: independently sample the output assigned to each input according to its channel row. Call the entire table U , independent of (X, Y) , and set $Z = f_U(X)$. Its marginal channel is the desired one. Conditioning reduces entropy, so $H(Y | Z) \geq H(Y | Z, U) = \mathbb{E}_U R(f_U) \geq R_k^*$. A deterministic optimum attains equality in the minimum. If a function uses fewer than k cells, refine it to k cells without increasing risk. $\square$

4 A sharp four-state obstruction

For L sufficiently large, set

$$w = e^{-L}, \quad \varepsilon = w^2, \quad \delta = \sqrt{\frac{\log 2}{L}}, \quad c = \frac{1 - 2w}{2}. \quad (6)$$

The source and binary-label posteriors are

Input	A	C	D	B
Probability	c	w	w	c
$\mathbb{P}(Y = 1 X)$	ε	δ	$1 - \delta$	$1 - \varepsilon$

Every entry of the joint distribution is positive. Symmetry gives $\mathbb{P}(Y = 1) = 1/2$. Let $\rho_{\text{det}, L}$ and $\rho_{\text{stoch}, L}$ denote (3) and (5) for this law.

Theorem 4.1 (Sharp deterministic and stochastic growth prices). *The family (6) satisfies*

$$\rho_{\text{det}, L} \sim \frac{\sqrt{L}}{2\sqrt{\log 2}}, \quad \rho_{\text{stoch}, L} \sim \frac{\sqrt{L}}{4\sqrt{\log 2}}. \quad (7)$$

For all sufficiently large L , the unique flat optimal partitions, up to cell labels, are

$$\mathcal{P}_2^* = \{AC, DB\}, \quad \mathcal{P}_3^* = \{A, CD, B\}.$$

The deterministic formula also holds when every total risk in (3) is replaced by excess risk D .

We prove the deterministic assertion first; Section 5 proves the stochastic assertion. The theorem excludes a universal constant simultaneous guarantee even for an oracle structural controller. It is not a computational hardness theorem.

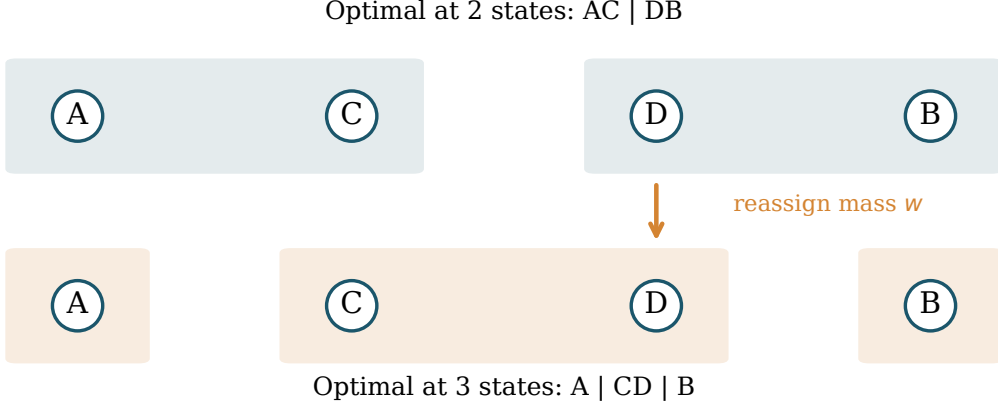


Figure 1: Schematic of the incompatible flat optima. At two states the rare inputs C, D join their neighboring common anchors. At three states they join each other. Splitting a cell of $\{AC, DB\}$ cannot produce $\{A, CD, B\}$. Reassigning one rare input after a split can do so. This illustration is schematic, not numerical evidence.

4.1 Exact reduction to six cluster costs

Write $h(t) = -t \log t - (1 - t) \log(1 - t)$ and define

$$J(\alpha, \beta; s, t) = (\alpha + \beta)h\left(\frac{\alpha s + \beta t}{\alpha + \beta}\right) - \alpha h(s) - \beta h(t).$$

This is the excess loss from merging two pure source states or two already formed cells with masses α, β and posterior means s, t . The cost of a triple is the cost of its first pair plus the cost of merging that pair's mean with the third point; substitution in J cancels the intermediate entropy exactly. Define

$$\begin{aligned} a &= J(c, w; \varepsilon, \delta), & b &= J(c, w; \varepsilon, 1 - \delta), \\ r &= 2w\{\log 2 - h(\delta)\}, & q &= 2c\{\log 2 - h(\varepsilon)\}, \\ t &= r + J(c, 2w; \varepsilon, 1/2), & u &= q + J(2c, w; 1/2, \delta). \end{aligned} \tag{8}$$

Here a is a near rare-anchor pair, b a far pair, r the rare pair, q the anchor pair, and t, u are triples with one and two anchors.

Lemma 4.2 (Complete partition enumeration). *The seven two-cell partition costs are $2a, 2b, q + r, t, t, u, u$. The six three-cell partition costs are a, a, b, b, r, q .*

Proof. A two-cell partition has sizes $2 + 2$ or $1 + 3$. The three $2 + 2$ choices are $AC|DB$, $AD|CB$, and $AB|CD$, with costs $2a, 2b, q + r$. The four $1 + 3$ choices isolate one of the two anchors (cost t) or one of the rare inputs (cost u). A three-cell partition merges precisely one pair; the six possible pairs have the listed costs. This covers every partition, independently of posterior ordering or any clustering algorithm. $\square$

4.2 Asymptotic estimates with explicit noise scaling

Lemma 4.3 (The six costs and irreducible risk). *Under (6),*

$$a \sim w\delta L, \quad b \sim wL, \quad r \sim 2w \log 2, \quad t \sim wL, \quad q \rightarrow \log 2, \quad u \rightarrow \log 2, \tag{9}$$

and $B_0 = 2ch(w^2) + 2wh(\delta) = o(w)$.

Proof. For $0 < x \leq 1/2$ the Taylor expansion of $-\log(1-x)$, or its geometric series integrated once, gives

$$h(x) = x \log(1/x) + x + O(x^2), \quad (10)$$

with an absolute remainder constant. We have $\delta \rightarrow 0$, $\delta L \rightarrow \infty$, and $\log(1/\delta) = O(\log L) = o(L)$.

Since $c + w = 1/2$, the mean of the near pair is $m = 2w\delta + (1 - 2w)w^2$. On the interval between $2w\delta$ and m , $h'(x) = \log((1-x)/x) = O(L)$. Thus $h(m) = h(2w\delta) + O(w^2L)$. Using $ch(w^2) = O(w^2L)$ and (10),

$$\begin{aligned} a &= w\delta\{L - \log(2\delta) + 1\} - wh(\delta) + O(w^2L) \\ &= w\delta(L - \log 2) + O(w\delta^2 + w^2L) \sim w\delta L. \end{aligned}$$

For the far pair, replace δ in the mean by $1 - \delta$ and retain $h(1 - \delta) = h(\delta)$. This gives

$$b = w(1 - \delta)\{L - \log(2(1 - \delta)) + 1\} - wh(\delta) + O(w^2L) \sim wL.$$

The formula for r in (8) gives its limit directly, as $h(\delta) \rightarrow 0$. The formula for q gives $q \rightarrow \log 2$.

The triple defining t has mass $M = c + 2w = (1 + 2w)/2$ and total positive-label mass $z = cw^2 + w = w + O(w^2)$. Equation (10) yields

$$Mh(z/M) = z\{\log(M/z) + 1\} + O(z^2/M) = w\{L - \log 2 + 1\} + O(w^2L).$$

Subtracting $ch(w^2) + 2wh(\delta)$ proves $t = wL + w\{1 - \log 2 - 2h(\delta)\} + O(w^2L) \sim wL$.

Finally $u \geq q$. Predicting $1/2$ after adding a rare state to the anchor pair bounds its extra Jensen cost by $wD_{\text{KL}}(\text{Ber}(\delta) \parallel \text{Ber}(1/2)) \leq w \log 2$. Hence $u - q \in [0, w \log 2]$ and $u \rightarrow \log 2$. The baseline statement follows because $h(w^2) = O(w^2L)$ and $h(\delta) \rightarrow 0$. $\square$

The noise scale is substantive: $\varepsilon = w^2$ makes the baseline $o(w)$. If one instead sets $w = \varepsilon^2$, a related excess-loss example exists, but its baseline dominates the relevant distortion. Such an example alone would not establish a total-risk ratio.

4.3 The optimal deterministic hierarchy

Define the total-risk ratios

$$\alpha_L = \frac{B_0 + a}{B_0 + r}, \quad \beta_L = \frac{B_0 + t}{B_0 + 2a}. \quad (11)$$

Proposition 4.4 (Exact eventual formula and Pareto frontier). *For all sufficiently large L , $D_2^* = 2a$, $D_3^* = r$ and*

$$\rho_{\text{det}, L} = \min\{\alpha_L, \beta_L\}. \quad (12)$$

Every deterministic hierarchy's vector of total-risk ratios at levels two and three is coordinate-wise at least one of $(1, \alpha_L)$ and $(\beta_L, 1)$, and both vectors are attainable.

Proof. Lemma 4.3 implies, eventually, $b > a > r$, $q > a$, $t > 2a$, and $t < q + r$, while u and $q + r$ are bounded away from zero. Thus the enumeration gives strict flat minima $2a$ and r .

If the three-cell partition merges CD , its possible two-cell costs are $t, t, q + r$. Its best vector is $(\beta_L, 1)$. If it merges a near pair, such as AC , it can reach $AC|DB$ at level two and attains $(1, \alpha_L)$. Any other three-cell merge has cost at least a , so its level-three ratio is at least α_L ; its level-two ratio is at least one by optimality of the benchmark. Both α_L and β_L exceed one eventually. These facts prove the Pareto statement and (12). $\square$

As $B_0 = o(w)$,

$$\alpha_L \sim \frac{\delta L}{2 \log 2}, \quad \beta_L \sim \frac{1}{2\delta}.$$

Our choice of δ makes both asymptotic to $\sqrt{L}/(2\sqrt{\log 2})$. This proves the deterministic part of Theorem 4.1. Setting $B_0 = 0$ in the ratio calculation proves the excess-loss assertion. More generally, if $\delta \rightarrow 0$ and $\log(1/\delta) = o(L)$, the same estimates give

$$\rho_{\text{det},L} = (1 + o(1)) \min \left\{ \frac{\delta L}{2 \log 2}, \frac{1}{2\delta} \right\}$$

whenever $\delta L \rightarrow \infty$. Balancing these two terms explains the selected family. This is sharp *within that family*; no matching worst-case upper rate for all four-state laws is asserted.

5 Randomized hierarchies and stochastic refinements

A random choice of a deterministic hierarchy is distinct from a stochastic encoder used on each input. We first solve the former problem exactly and then use it for a converse that covers the latter.

Proposition 5.1 (Randomized hierarchy value). *For sufficiently large L , the infimum over distributions on deterministic hierarchies of $\max_{k=2,3} \mathbb{E}R(\mathcal{H}_k)/R_k^*$ equals*

$$\rho_{\text{mix},L} = 1 + \frac{(\alpha_L - 1)(\beta_L - 1)}{\alpha_L + \beta_L - 2} \sim \frac{\sqrt{L}}{4\sqrt{\log 2}}. \quad (13)$$

Proof. Proposition 4.4 permits replacing each hierarchy by one of the two attainable Pareto vectors without increasing either expected coordinate. Put probability λ on $(1, \alpha_L)$ and $1 - \lambda$ on $(\beta_L, 1)$. The coordinates are $1 + (1 - \lambda)(\beta_L - 1)$ and $1 + \lambda(\alpha_L - 1)$. Their maximum is minimized at equality, $\lambda = (\beta_L - 1)/(\alpha_L + \beta_L - 2)$, giving (13). The asymptotic follows from $\alpha_L \sim \beta_L \sim \sqrt{L}/(2\sqrt{\log 2})$. $\square$

If instead the criterion is $\mathbb{E} \max_k R(\mathcal{H}_k)/R_k^*$, randomizing cannot improve the best deterministic value. Equation (13) uses the maximum of expected risks, which is the stronger lower-bound setting needed next.

Lemma 5.2 (Random-function converse for stochastic refinement). *Every pair of channels in (4) satisfies*

$$\max_{k=2,3} \frac{H(Y \mid Z_k)}{R_k^*} \geq \rho_{\text{mix},L}.$$

Proof. Independently of (X, Y) , draw a table F assigning a Z_3 value to each x according to $K_3(\cdot \mid x)$, with independent entries. Draw a second table G_0 assigning a Z_2 value to each z_3 according to $G(\cdot \mid z_3)$, also independently. The finite random variable $U = (F, G_0)$ satisfies $Z_3 = F(X)$ and $Z_2 = G_0(F(X))$ and reproduces the prescribed joint law. For each fixed U these functions form nested partitions with at most three and two cells.

These partitions can be refined to exactly three and two cells while remaining nested and without increasing either risk. To see this, first split fine cells until there are three. Each is still contained in an original coarse cell. If there were two coarse cells, retain them; if there was one, divide the three fine cells into two nonempty groups. The resulting partitions extend to a complete four-state hierarchy by adding the singleton and one-cell levels. Call this hierarchy $\mathcal{H}^U$. For each k , conditioning and refinement therefore give

$$H(Y \mid Z_k) \geq H(Y \mid Z_k, U) \geq \mathbb{E}_U R(\mathcal{H}_k^U).$$

Apply the definition of $\rho_{\text{mix},L}$ to the distribution of $\mathcal{H}^U$. The random table is used only in a converse by revealing more information; it is not assumed to be free information available to the original encoder's predictor. $\square$

Proposition 5.3 (Matching stochastic construction). *A three-state encoder and deterministic two-state coarsening attain*

$$\max_{k=2,3} \frac{H(Y | Z_k)}{R_k^*} \sim \frac{\sqrt{L}}{4\sqrt{\log 2}}.$$

Proof. For a fixed $\theta \in (0, 1)$, use three outputs 0, m , 1 and the channel

	0	m	1
A	1	0	0
C	0	1	0
D	0	θ	$1 - \theta$
B	0	0	1

Coarsen 0, m to one output and retain 1 as the other. This directly satisfies the output budgets; it does not use a public mixing label.

We calculate the leading entropies. The three-state middle output has mass $w(1 + \theta)$ and posterior $[\delta + \theta(1 - \delta)]/(1 + \theta)$, so its risk contribution is $O(w)$. Output 0 contributes $ch(w^2) = O(w^2L)$. Output 1 has mass $c + (1 - \theta)w \rightarrow 1/2$ and failure-label mass $cw^2 + (1 - \theta)w\delta \sim (1 - \theta)w\delta$. Equation (10), together with $\log(1/\delta) = o(L)$, shows that this contribution is $(1 + o(1))(1 - \theta)w\delta L$. As $w = o(w\delta L)$,

$$H(Y | Z_3) \sim (1 - \theta)w\delta L.$$

At level two, the merged low output has mass $c + w(1 + \theta) \rightarrow 1/2$ and positive-label mass $cw^2 + w\delta + \theta w(1 - \delta) \sim \theta w$. Its entropy contribution is $(1 + o(1))\theta wL$; the other output contributes only $O(w\delta L)$. Hence $H(Y | Z_2) \sim \theta wL$. By Lemma 4.3, $R_2^* \sim 2w\delta L$ and $R_3^* \sim 2w \log 2$. The two ratios are consequently asymptotic to

$$\theta \frac{\sqrt{L}}{2\sqrt{\log 2}} \quad \text{and} \quad (1 - \theta) \frac{\sqrt{L}}{2\sqrt{\log 2}}.$$

Choose $\theta = 1/2$. Combining this feasible upper bound with Lemma 5.2 and (13) proves both the proposition and the stochastic part of Theorem 4.1. $\square$

6 Regularity, robustness, and the cost of reassignment

6.1 A finite bound when source states cannot be arbitrarily rare

The next bound helps identify which assumptions drive the counterexample. It allows any finite target alphabet and even zero target probabilities. Its constant is not claimed to be optimal.

Theorem 6.1 (Four-state source-mass bound). *If $|\mathcal{X}| = 4$ and $p_x \geq \gamma > 0$, there is a deterministic hierarchy satisfying, simultaneously for $k = 2, 3$,*

$$D(\mathcal{H}_k) \leq 10\{2 + \log(1/\gamma)\}D_k^*, \quad R(\mathcal{H}_k) \leq 10\{2 + \log(1/\gamma)\}R_k^*. \quad (14)$$

The inequalities include zero-optimum cases without division.

Proof. We prove the divergence comparison and hierarchy construction explicitly. For probability vectors P, Q with $P_j \leq MQ_j$ for every coordinate, where $M \geq 1$, define $\text{He}^2(P, Q) = \sum_j (\sqrt{P_j} - \sqrt{Q_j})^2$. Then

$$\text{He}^2(P, Q) \leq D_{\text{KL}}(P \| Q) \leq (2 + \log M) \text{He}^2(P, Q). \quad (15)$$

For completeness, put $f(t) = t \log t - t + 1$ and $z = \sqrt{t}$. The lower bound follows from $f(z^2) - (z - 1)^2 = 2z(z \log z - z + 1) \geq 0$, interpreted continuously at zero. For the upper bound let $g(z) = f(z^2)$. We have $g(1) = g'(1) = 0$ and $g''(z) = 4 + 4 \log z$. On $[1, \sqrt{M}]$ this is at most $4 + 2 \log M$, and on $(0, 1]$ it is at most 4. Integrating the second derivative from one, in the appropriate direction, gives $g(z) \leq (2 + \log M)(z - 1)^2$; continuity handles zero. Multiply by Q_j and sum, using $\sum_j Q_j f(P_j/Q_j) = D_{\text{KL}}(P \| Q)$. Coordinates with $Q_j = 0$ also have $P_j = 0$ and contribute zero.

Let $v_x = (\sqrt{P_x(y)})_y$ and let $V(\mathcal{P})$ be the weighted Euclidean within-cell sum of squares of the v_x . In a cell of mass p_C , its normalized weights $a_x = p_x/p_C$ satisfy $a_x \geq \gamma$. The mixture P_C obeys $P_x(y)/P_C(y) \leq 1/a_x \leq 1/\gamma$. Let $\bar{v} = \sum_{x \in C} a_x v_x$ and $s = \sqrt{P_C}$ coordinatewise. Concavity of the square root gives $s \geq \bar{v}$ coordinatewise; also $\|v_x\| = \|s\| = 1$. Therefore

$$\sum_x a_x \|v_x - s\|^2 = 2 - 2\langle \bar{v}, s \rangle \leq 2(1 - \|\bar{v}\|^2) = 2 \sum_x a_x \|v_x - \bar{v}\|^2.$$

The reverse inequality without the factor two holds because $\bar{v}$ minimizes Euclidean squared error. Combining this with (15) cell by cell proves

$$V(\mathcal{P}) \leq D(\mathcal{P}) \leq 2\{2 + \log(1/\gamma)\}V(\mathcal{P}). \quad (16)$$

It remains to construct a five-approximate Euclidean hierarchy on four weighted points; their dimension is immaterial. Choose a pair (i, j) of minimum pair cost

$$s_* = \min_{i < j} \frac{p_i p_j}{p_i + p_j} \|v_i - v_j\|^2,$$

and merge it at level three, which is optimal for V_3^* . For a cluster C of m points and total mass W , its squared error is

$$V(C) = \frac{1}{W} \sum_{i < j \in C} p_i p_j \|v_i - v_j\|^2 \geq \frac{s_*}{W} \sum_{i < j \in C} (p_i + p_j) = (m - 1)s_*.$$

Consequently $V_2^* \geq 2s_*$. If an optimal two-cell partition contains the chosen pair in one cell, it already coarsens the three-cell partition. Otherwise move the lighter point of the pair, say $p_i \leq p_j$, into j 's optimal cell. Using that cell's old centroid b as a candidate centroid bounds the new cost by

$$\begin{aligned} V_2^* + p_i \|v_i - b\|^2 &\leq V_2^* + 2p_i \|v_i - v_j\|^2 + 2p_i \|v_j - b\|^2 \\ &\leq 3V_2^* + 4s_* \leq 5V_2^*. \end{aligned}$$

The term removed from i 's old cell was nonnegative and was dropped. We used $p_i \leq 2p_i p_j / (p_i + p_j)$ and $p_i \|v_j - b\|^2 \leq p_j \|v_j - b\|^2 \leq V_2^*$. Reoptimizing centroids can only decrease the cost. The resulting cells are unions of the three chosen cells. If moving i emptied its old cell, there is one cell; split it into two unions of the three cells, which again cannot increase squared error. This proves the simultaneous factors five and one for V .

Let $C_\gamma = 2 + \log(1/\gamma)$. Since $V_k^* \leq D_k^*$, (16) gives $D(\mathcal{H}_k) \leq 2C_\gamma \cdot 5V_k^* \leq 10C_\gamma D_k^*$. Adding the nonnegative common baseline B_0 preserves the same factor for total risks, proving (14). $\square$

In the constructed family $\gamma = w = e^{-L}$, so the lower price is of order $\sqrt{\log(1/\gamma)}$, whereas Theorem 6.1 is of order $\log(1/\gamma)$. Closing this gap for the worst four-state instance is open here. The upper bound nevertheless proves that, with four inputs, arbitrarily rare source states are necessary for unbounded price; extreme conditional label probabilities alone are insufficient.

6.2 Minimal cardinalities and perturbation robustness

Proposition 6.2 (Minimality and open-set robustness). *Four is the smallest source cardinality at which the deterministic all-capacity hierarchy factor can exceed one. Two is the smallest nontrivial target cardinality. For every finite M , a nonempty open set of full-support four-by-two joint laws has both deterministic and stochastic hierarchy price greater than M .*

Proof. For at most three source states, only the two-cell level can be nontrivial; any optimum there extends to the singleton and one-cell partitions. A one-state target has zero prediction risk at every representation. For robustness choose L large enough that the mixed-hierarchy value in (13) exceeds $2M$. There are finitely many deterministic hierarchies. All their risk coordinates and positive flat benchmarks are continuous functions of the full-support joint law. Their normalized coordinate vectors therefore vary uniformly continuously locally. The minimum over mixtures of the maximum of two expected coordinates changes by at most the largest change in any coordinate. Thus it remains above M on some open neighborhood. The random-function converse applies to every law on the same finite alphabets and bounds the stochastic and deterministic prices below by this mixture value. $\square$

6.3 A vanishing amount of reassignment restores both flat optima

Let Z_2^* and Z_3^* identify $\{AC, DB\}$ and $\{A, CD, B\}$. A label-invariant measure of incompatibility is the minimum error in recovering the old coarse label from the new fine label:

$$\text{rec}(Z_2^* | Z_3^*) = \min_g \mathbb{P}\{g(Z_3^*) \neq Z_2^*\}. \quad (17)$$

Proposition 6.3 (Exact reassignment cost). *For the family (6),*

$$\text{rec}(Z_2^* | Z_3^*) = w, \quad H(Z_2^* | Z_3^*) = 2w \log 2.$$

Splitting one cell and reassigning one input of mass w connects the optimal two- and three-cell partitions.

Proof. The singleton fine cells A, B determine the old label exactly. In the fine cell CD , of mass $2w$, the two old labels occur with equal conditional probability. Any deterministic recovery map errs on at least one mass- w input; assigning the cell to either old label attains that error. Its conditional entropy is $\log 2$, giving the entropy formula. To realize the change, split AC into A and C , leaving DB as the third cell; then move D from DB into C . The result is $A|CD|B$ and the moved source mass is exactly w . $\square$

This is a statement about reassignment in the finite representation model. It does not prescribe the parameter cost of moving a neuron’s activation region. It shows why measuring only state count or retained information misses a separate resource: the ability to reorganize earlier assignments.

7 An extension to convex Bayes-risk generators

The logarithmic example has a broader counterpart for excess prediction loss. This extension concerns the Jensen gap of a convex generator; it does not assert the total-risk conclusion proved for logarithmic loss. In particular, additive affine terms do not affect its distortion.

Let $\phi : [0, 1] \rightarrow \mathbb{R}$ be continuous and strictly convex, and differentiable on $(0, 1)$. For a finite source with masses $p_x > 0$ and binary-label posteriors $\eta_x \in (0, 1)$, define

$$D_\phi(\mathcal{P}) = \sum_{C \in \mathcal{P}} \left\{ \sum_{x \in C} p_x \phi(\eta_x) - p_C \phi(\bar{\eta}_C) \right\}, \quad p_C = \sum_{x \in C} p_x, \quad \bar{\eta}_C = p_C^{-1} \sum_{x \in C} p_x \eta_x. \quad (18)$$

This is the excess Bayes risk when the concave Bayes-risk function is $-\phi$, up to an affine term, as in the convex representation of proper scoring rules [5, Theorem 1]. The identity follows by subtracting the Bayes risk with the full source from that with the partition label. Equivalently, putting

$$d_\phi(z \| x) = \phi(z) - \phi(x) - \phi'(x)(z - x), \quad (19)$$

the distortion in (18) equals $\sum_C \sum_{x \in C} p_x d_\phi(\eta_x \| \bar{\eta}_C)$, because the linear terms sum to zero within every cell.

Write $D_{\phi,k}^*$ for the minimum of (18) over partitions with k nonempty cells. For a four-state source, let

$$\rho_\phi = \min_{\mathcal{H}} \max_{k \in \{2,3\}} \frac{D_\phi(\mathcal{H}_k)}{D_{\phi,k}^*}, \quad (20)$$

where $\mathcal{H}$ ranges over complete binary-merger hierarchies. The constructions below have four distinct posteriors, so both denominators are strictly positive.

Theorem 7.1 (Divergent endpoint slope). *Suppose, in addition, that $\phi(x) = \phi(1 - x)$ and $\phi'(x) \rightarrow -\infty$ as $x \downarrow 0$. Put*

$$C = \phi(0) - \phi(1/2) > 0, \quad s_\varepsilon = -\phi'(\varepsilon), \quad \delta_\varepsilon = \sqrt{C/s_\varepsilon}.$$

There is a choice of positive $w_\varepsilon \leq \varepsilon^2$ such that the four-state source with masses

$$(p_A, p_C, p_D, p_B) = (c, w, w, c), \quad c = (1 - 2w)/2,$$

and posteriors

$$(\eta_A, \eta_C, \eta_D, \eta_B) = (\varepsilon, \delta, 1 - \delta, 1 - \varepsilon), \quad w = w_\varepsilon, \quad \delta = \delta_\varepsilon,$$

satisfies

$$\rho_\phi \sim \frac{\sqrt{s_\varepsilon}}{2\sqrt{C}} \rightarrow \infty \quad (\varepsilon \downarrow 0). \quad (21)$$

All source and conditional label probabilities are positive for sufficiently small ε .

Proof. We first justify a choice of w that makes the required expansions valid even when ϕ has no uniform second-derivative bound near zero. For positive weights α, β , define the weighted Jensen gap

$$G(\alpha, \beta; x, z) = \alpha\phi(x) + \beta\phi(z) - (\alpha + \beta)\phi\left(\frac{\alpha x + \beta z}{\alpha + \beta}\right).$$

If $m = (\alpha x + \beta z)/(\alpha + \beta)$, direct cancellation gives

$$G(\alpha, \beta; x, z) = \beta d_\phi(z \| x) - (\alpha + \beta) d_\phi(m \| x). \quad (22)$$

Fix an interior ε . Differentiability at ε means $d_\phi(\varepsilon + h|\varepsilon) = o(|h|)$ as $h \rightarrow 0$. For $\alpha = c = (1 - 2w)/2$, $\beta = kw$, $k \in \{1, 2\}$, and $z \in [0, 1]$, one has $|m - \varepsilon| \leq 4w$ for sufficiently small w . Thus the remainder in (22) is $o(w)$ uniformly over these k and z , with ε fixed. Since $s_\varepsilon \rightarrow \infty$, we may choose, separately for each sufficiently small ε , a positive $w = w_\varepsilon \leq \varepsilon^2$ for which

$$0 \leq kwd_\phi(z|\varepsilon) - G(c, kw; \varepsilon, z) \leq w/s_\varepsilon \quad (k \in \{1, 2\}, z \in [0, 1]). \quad (23)$$

The uniform statement follows directly from the definition of differentiability on the shrinking interval $[\varepsilon - 4w, \varepsilon + 4w]$; w can be reduced further so this interval is interior. This is an existence choice of the rare mass, rather than an assumption of uniform differentiability as ε changes.

There are two useful preliminary limits. The supporting-line inequality at ε gives

$$0 \leq \varepsilon s_\varepsilon \leq 2\{\phi(\varepsilon/2) - \phi(\varepsilon)\} \rightarrow 0.$$

It follows that $ws_\varepsilon \rightarrow 0$ and $\varepsilon/\delta \rightarrow 0$, while $\delta \rightarrow 0$ and $\delta s_\varepsilon = \sqrt{Cs_\varepsilon} \rightarrow \infty$. Strict convexity and symmetry imply $C > 0$.

For clarity, write $s = s_\varepsilon$ and denote the six distinct cluster costs by

$$\begin{aligned} a &= G(c, w; \varepsilon, \delta), & b &= G(c, w; \varepsilon, 1 - \delta), \\ r &= 2w\{\phi(\delta) - \phi(1/2)\}, & q &= 2c\{\phi(\varepsilon) - \phi(1/2)\}, \\ t &= r + G(c, 2w; \varepsilon, 1/2), & u &= q + G(2c, w; 1/2, \delta). \end{aligned}$$

Here a, b are the near and far rare-anchor pair costs, r, q are the rare-pair and anchor-pair costs, and t, u are the triple costs with respectively one and two anchors. These expressions follow by first merging the pair inside a triple and then merging its centroid with the remaining point; expanding both sides proves this aggregation identity exactly.

By (23),

$$a = w\{\phi(\delta) - \phi(\varepsilon) + s(\delta - \varepsilon)\} + o(w) \sim w\delta s, \quad b \sim ws.$$

The first equivalence uses boundedness of ϕ on $[0, 1]$, $\varepsilon s \rightarrow 0$, and $\delta s \rightarrow \infty$; the second uses the same bounds and $\delta \rightarrow 0$. Continuity at the endpoints gives $r \sim 2wC$ and $q \rightarrow C$. Also,

$$t = 2w\{\phi(\delta) - \phi(\varepsilon)\} + ws(1 - 2\varepsilon) + o(w) \sim ws.$$

Finally, symmetry and differentiability imply $\phi'(1/2) = 0$. Applying (22) at $1/2$ and using convexity yields

$$0 \leq u - q \leq wd_\phi(\delta|1/2) = w\{\phi(\delta) - \phi(1/2)\} \leq wC.$$

Consequently $u \rightarrow C$.

We now check all possible partitions rather than assuming that a particular merger algorithm is optimal. The seven two-cell costs are $2a, 2b, q + r, t, t, u, u$, and the six three-cell costs are a, a, b, b, r, q . The limits just proved show that, for sufficiently small ε ,

$$D_{\phi,2}^* = 2a, \quad D_{\phi,3}^* = r.$$

They also show $b > a > r$, $q > a$, $t > 2a$, and $t < q + r$. If a hierarchy merges the rare pair at level three, its possible level-two costs are $t, t, q + r$, so its best factor is $t/(2a)$. If it merges a near rare-anchor pair, it can attain the optimal level-two partition and has best factor a/r . Every other level-three pair has cost at least a , hence cannot improve on the latter factor. We have therefore proved the exact eventual formula

$$\rho_\phi = \min\{a/r, t/(2a)\}.$$

Its two entries satisfy

$$\frac{a}{r} \sim \frac{\delta s}{2C} = \frac{\sqrt{s}}{2\sqrt{C}}, \quad \frac{t}{2a} \sim \frac{1}{2\delta} = \frac{\sqrt{s}}{2\sqrt{C}}.$$

This proves (21). $\square$

The choice of w matters. For a general differentiable generator, setting $w = \varepsilon^2$ without the further reduction in (23) need not justify the Taylor remainder. The logarithmic constructions admit explicit estimates and therefore do not require this adaptive choice. Moreover, Theorem 7.1 is a sufficient condition for an unbounded hierarchy factor; it is not a characterization of all generators with this property.

Proposition 7.2 (A comparison on intervals of bounded curvature). *Let all posteriors lie in a fixed compact interval $J \subset (0, 1)$, and suppose ϕ is twice continuously differentiable there with $0 < m \leq \phi''(x) \leq M < \infty$. Define weighted squared-error distortion by*

$$E(\mathcal{P}) = \sum_{C \in \mathcal{P}} \sum_{x \in C} p_x(\eta_x - \bar{\eta}_C)^2.$$

Then every partition satisfies

$$\frac{m}{2} E(\mathcal{P}) \leq D_\phi(\mathcal{P}) \leq \frac{M}{2} E(\mathcal{P}). \quad (24)$$

For any collection of cardinalities whose optimal distortions are positive, the minimum simultaneous hierarchy factors obey

$$\frac{m}{M} \rho_E \leq \rho_\phi \leq \frac{M}{m} \rho_E.$$

Proof. The integral form of Taylor's theorem is

$$d_\phi(z\|x) = (z - x)^2 \int_0^1 (1 - t) \phi''(x + t(z - x)) dt.$$

Both x, z and their intervening segment belong to J , so this expression lies between $m(z - x)^2/2$ and $M(z - x)^2/2$. Summing at the cell centroids proves (24). Minimizing its two bounds over all k -cell partitions gives the corresponding bounds on optimal distortions. Dividing the partition bounds by these optimal bounds, taking the maximum over cardinalities, and then the minimum over hierarchies proves the last assertion. $\square$

This comparison transfers hierarchy guarantees between the two losses up to the fixed condition ratio M/m . It does not supply an independent numerical constant for squared-error hierarchies.

8 Reproducible computational investigations

The experiments evaluate the mathematical objects directly. No neural training run is needed to test a theorem about finite encoding channels, and no empirical neural-network performance is inferred from these calculations. An exploratory random search suggested the rare-anchor pattern; its original script, seed, and rerun result are preserved separately. The following calculations were specified after choosing the analytic family.

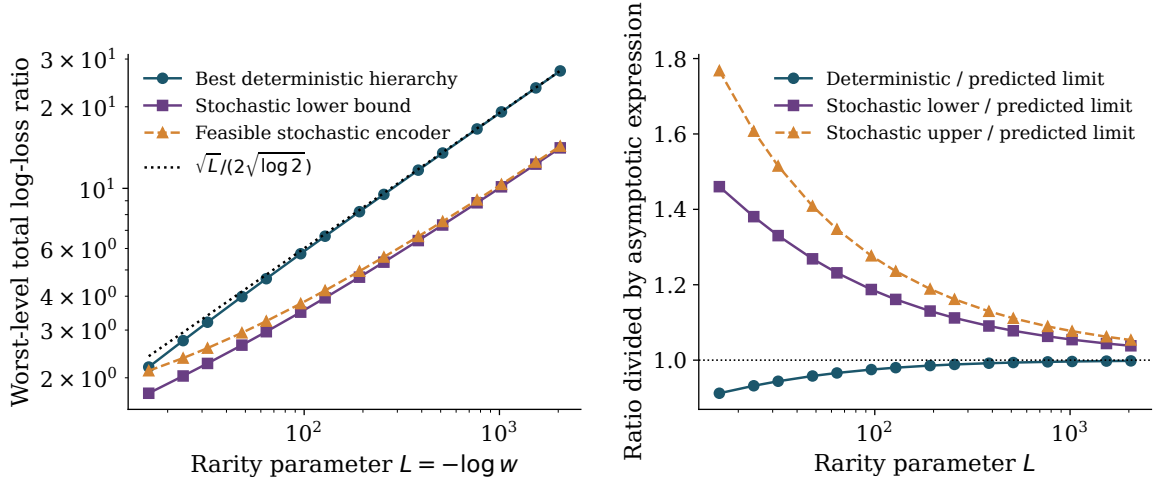


Figure 2: Computed hierarchy prices and stochastic bounds from exhaustive high-precision evaluation. Left: the best deterministic total-risk ratio, the stochastic lower bound from random hierarchies, and the feasible three-state stochastic encoder. Right: division by the respective predicted asymptotic expressions; both stochastic bounds converge toward the same constant. The stochastic upper curve is a feasible construction, not a numerically certified global optimum.

8.1 Exhaustive evaluation and precision

The program `code/quartet_experiment.py` generates all fifteen set partitions of four inputs, all seven two-cell partitions, all six three-cell partitions, and all eighteen compatible three-to-two chains. It evaluates their entropies with arbitrary precision, takes the exact finite minimum numerically, and checks the formula in Proposition 4.4. These are exhaustive finite comparisons, not local optimization over selected starting points.

The scaling investigation uses fifteen values:

$$L \in \{16, 24, 32, 48, 64, 96, 128, 192, 256, 384, 512, 768, 1024, 1536, 2048\}.$$

Decimal precision is $\lceil 2L/\log 10 \rceil + 90$, with a minimum of eighty, so the small posterior e^{-2L} is represented before subtracting entropies. The parameters, baseline, partition costs, hierarchy scores, and stochastic construction risks are saved as sixty-digit decimal strings. Four representative cases, including $L = 2048$, are recalculated with seventy additional digits; the saved deterministic prices, mixture values, stochastic bounds, and construction risks agree to all sixty stored digits. Formula comparisons use relative tolerance 10^{-40} . This is numerical validation, not a formal arithmetic certificate or a replacement for the proofs.

For random hierarchies, the program constructs the risk vectors of all eighteen chains and checks the diagonal intersections of every pair of vectors, as well as all vertices. The minimum of the maximum coordinate on a convex polygon occurs at a vertex or a boundary intersection with the diagonal, so this finite calculation recovers the mixture value. It agrees with (13). The explicitly specified stochastic encoder with $\theta = 1/2$ supplies an upper bound on ρ_{stoch} ; the mixture value supplies a lower bound. We do not label the feasible stochastic value as a finite- L global optimum.

8.2 Scale sensitivity and an unfavorable loss comparison

At $L = 256$ we vary $\delta/\sqrt{\log(2)/L} \in \{0.35, 0.5, 0.65, 0.8, 1, 1.2, 1.5, 2, 2.5, 3\}$. The two competing excess-loss ratios move in opposite directions; their minimum displays the predicted

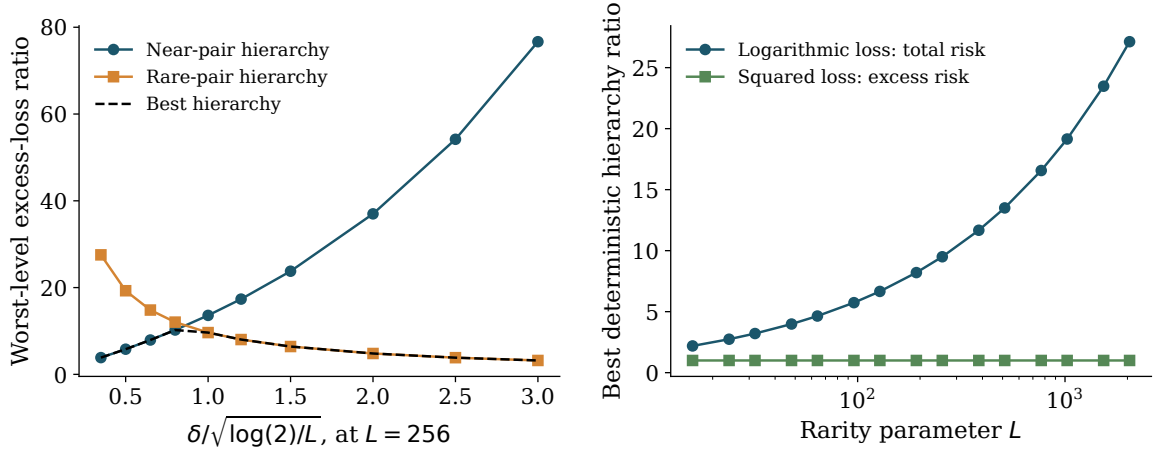


Figure 3: Computed sensitivity and loss comparison. Left: changing the intermediate posterior scale at $L = 256$ trades the cost of near-pair and rare-pair hierarchies; the dashed line is the best excess-loss ratio. Right: the same source family has growing total-log-loss hierarchy price, while its squared-loss excess hierarchy price equals one on all evaluated cases. The two curves use the explicitly indicated loss criteria.

tradeoff. The finite- L maximizing multiplier need not equal the asymptotic choice one, since $h(\delta)$ corrections decay slowly.

The same distribution family is also evaluated under squared prediction loss. Its excess Bayes risk is weighted posterior squared error. Every sampled L has an exactly compatible pair of flat optimal partitions and computed hierarchy factor one. This control shows that the construction is not a loss-independent impossibility; it reflects logarithmic loss’s sensitivity near the probability boundary. The bounded-curvature comparison in Proposition 7.2 makes this distinction mathematical.

L	$\log_{10} w$	ρ_{det}	Stochastic lower	Feasible upper	B_0/D_3^*
16	-6.9	2.19133	1.75386	2.12340	2.81549
64	-27.8	4.64119	2.95855	3.23657	0.92962
256	-111.2	9.49943	5.34252	5.57974	0.41840
1024	-444.7	19.14974	10.13188	10.34765	0.21067
2048	-889.4	27.12538	14.10659	14.31553	0.15253

Table 1: Selected computed values. The deterministic column uses total risk. The stochastic columns are a variational lower bound evaluated numerically and the risk of the explicit encoder, respectively. The underlying values are saved with sixty significant decimal digits.

The absolute losses approach zero as L grows, while their ratios diverge. This is a failure of a uniform multiplicative guarantee near a highly predictable target, not a lower bound on fixed-scale absolute damage. The source masses become extremely small: at $L = 2048$, the rare mass is about 10^{-889} . Ordinary random sampling would not expose such states at practical sample sizes. The computation uses a known finite population law to make the theorem inspectable; it does not establish practical learnability of the rare states.

9 Interpretation and remaining questions

The original question concerned information and self-organizing neural structure. The result suggests that one must specify *which structural operations information is allowed to guide*. In this finite model, the controller can know the exact task distribution, evaluate every partition, and use arbitrary computation. A restriction to nested refinements still prevents a uniform approximation guarantee. The obstruction is therefore not repaired by a more accurate entropy estimator or a more sophisticated split score alone.

The conclusions also separate two resources often grouped under capacity. A third state makes the independently optimal representation better, but reaching it may require moving information across an old boundary. Stochastic routing reduces the asymptotic price by exactly two in the constructed family. Full reassignment of a mass- w input restores both flat optima. State count, stochasticity, and reassignment are distinct resources in this analysis.

Several limitations delimit the result. The model counts discrete representation states, not parameters, real-valued feature coordinates, neurons, or energy. It permits Bayes-optimal decoding and assumes a known joint law. It concerns the two specific intermediate budgets, so a process that skips a budget is outside the simultaneous guarantee. It does not exclude effective growth rules for restricted distributions, finite-sample objectives, additive accuracy criteria, or architectures that naturally reorganize representations. The full-support and open-neighborhood results remove exact zeros and exact symmetry as requirements for a given finite gap, but the size of that neighborhood need not be uniform in L .

The source-mass upper bound leaves a concrete quantitative problem: is the worst four-state price of order $\sqrt{\log(1/\gamma)}$, of order $\log(1/\gamma)$, or intermediate? The endpoint-slope theorem gives a sufficient condition for excess-loss obstruction, but no necessary and sufficient classification of scoring rules. A broader research direction is to characterize how a limited reassignment budget controls the price of structural growth. These questions are not needed for the established theorems and remain unresolved in this work.

References

- [1] Anna Arutyunova and Heiko Röglin. The price of hierarchical clustering. In *30th Annual European Symposium on Algorithms (ESA 2022)*, volume 244 of *Leibniz International Proceedings in Informatics*, pages 10:1–10:14, 2022. doi: 10.4230/LIPIcs.ESA.2022.10. URL <https://doi.org/10.4230/LIPIcs.ESA.2022.10>.
- [2] Arindam Banerjee, Srujana Merugu, Inderjit S. Dhillon, and Joydeep Ghosh. Clustering with Bregman divergences. *Journal of Machine Learning Research*, 6:1705–1749, 2005. URL <https://jmlr.org/papers/volume6/banerjee05b/banerjee05b.pdf>.
- [3] Hippolyte Charvin, Nicola Catenacci Volpi, and Daniel Polani. Exact and soft successive refinement of the information bottleneck. *Entropy*, 25(9):1355, 2023. doi: 10.3390/e25091355. URL <https://doi.org/10.3390/e25091355>.
- [4] Kamalika Chaudhuri and Andrew McGregor. Finding metric structure in information theoretic clustering. In *21st Annual Conference on Learning Theory (COLT)*, pages 391–402, 2008. URL <https://cseweb.ucsd.edu/~kamalika/pubs/cm08.pdf>.
- [5] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437. URL <https://sites.stat.washington.edu/people/raftery/Research/PDF/Gneiting2007jasa.pdf>.
- [6] Assaf Kartowsky and Ido Tal. Greedy-merge degrading has optimal power-law. *IEEE Transactions on Information Theory*, 65(2):917–934, 2019. doi: 10.1109/TIT.2018.2879802. URL <https://webee.technion.ac.il/people/idotai/papers/journal/DegradingUpgrading.pdf>.
- [7] Noam Slonim and Naftali Tishby. Agglomerative information bottleneck. URL <https://www.cise.ufl.edu/~anand/sp06/ib.pdf>. Undated extended manuscript, Section 3.1; hosted in a University of Florida course archive.
- [8] Noam Slonim and Naftali Tishby. Agglomerative information bottleneck. In *Advances in Neural Information Processing Systems*, volume 12, pages 617–623, 1999. URL <https://proceedings.neurips.cc/paper/1999/hash/be3e9d3f7d70537357c67bb3f4086846-Abstract.html>.